Alzheimer's Disease Detection Using Deep Learning Method

A Project Report Submitted to Cotton University in Partial Fulfillment of the Requirements for the Degree of

Master of Computer Application (MCA)

In the Department of Computer Science and Information Technology

By

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4th Semester



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JUNE 2026

CERTIFICATE

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This is to certify that the project entitled "Alzheimer's Disease Detection Using Deep Learning Method" submitted by Pradumna Kalita for the award of the degree of Master of Computer Application (MCA) in the Department of Computer Science and Information Technology is the outcome of a bona fide project work under my/our supervision. This work has not been submitted previously for any other degree of this or any other University. It is further certified that the candidate has complied with all the formalities as per the requirements of Cotton University, Guwahati-01. I/We recommend that the project report may be accepted in partial fulfillment of the requirements for the degree of MCA of this University.

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TO WHOM IT MAY CONCERN

This is to certify that **Pradumna Kalita**, a Master of Computer Application (MCA) 4th semester student of **Cotton University**, bearing Roll No./Registration No. **MCA2465004**, has successfully completed an Internship under my supervision in the **Department of Information Technology, Gauhati University**, during the period from **January, 2026** to **May, 2026**.

During the course of the internship, the student worked on the topic entitled "**Alzheimer's Disease Detection using Deep Learning Method**"

The student carried out the assigned work sincerely and diligently, and also demonstrated satisfactory understanding, dedication, and professional conduct throughout the internship period.

To the best of my knowledge, the work undertaken during this internship was completed successfully under my supervision and guidance.

I wish him success in future academic and professional endeavors.

Regard.

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DECLARATION

I, Pradumna Kalita, bearing MCA Enrollment No.:MCA2465004 hereby declare that the subject matter of the project entitled "Alzheimer's Disease Detection Using Deep Learning Method" is the record of work done by me under the guidance of Dr. Sangita Barauh, Department of Computer Science and Information Technology, Cotton University, Guwahati-01, Assam. I further declare that the contents of this project report did not form the basis for the award of any degree to me or to anybody else to the best of my knowledge. The report has not been submitted to any other University or Institution. This report is being submitted to Cotton University, Guwahati-01 for the degree of MCA in the Department of Computer Science and Information Technology.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AD	Alzheimer’s Disease
CN	Cognitive Normal
MRI	Magnetic Resonance Imaging
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
ResNet	Residual Neural Network
ReLU	Rectified Linear Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
ADNI	Alzheimer's Disease Neuroimaging Initiative
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
MCI	Mild Cognitive Impairment
AUC	Area Under Curve
F1-score	Harmonic Mean of Precision and Recall
PET	Positron Emission Tomography

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ABSTRACT

Alzheimer's disease (AD) is a progressive neurodegenerative condition that affects a person's cognitive functions such as memory, thinking ability and makes an impact on daily activities. There is still no cure of the disease so early detection is very important for timely treatment of the disease to slow down the progression. This study proposes a method of classifying Alzheimer's Disease (AD) and Cognitively Normal (CN) patients using a deep learning model on MRI data.

For this study MRI data collected from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. The data went through various cleaning and preprocessing training techniques before training and evaluation. A ResNet18 model was used as a pretrained model in transfer learning to classify AD and CN patients into two classes. Multiple experiments were conducted to analyze the effect of data augmentation, freezing layers, fine-tuning, learning rates, weight decay and threshold tuning.

Experimental results indicated that data augmentation helped to improve the model performance. Specifically, the best-performing model yielded an accuracy score of 88.46% and an F1-score of 72.73% on an augmented dataset for a fixed ResNet18 backbone network. Furthermore, subject-based evaluation was conducted to generate clinically relevant outcomes. Additionally, Grad-CAM was utilized to identify important brain regions.

Therefore, the findings above have shown that transfer learning along with data augmentation is highly effective in classifying Alzheimer's Disease using MRI scans.

CHAPTER 1: INTRODUCTION

1.1 Alzheimer's Disease Overview

Alzheimer's disease is a type of dementia that is affected by decline of a person's brain memory, thinking ability and their behaviour. Which becomes serious enough to interfere with normal life. The biggest known risk factor is increasing age. The majority of people who have Alzheimer's were 65 or more older. According to the World Health Organization (WHO) report in 2021, 57 millions people worldwide are living with Alzheimer's and other types of dementia. In India the Indian Journal of Public Health reports that according to the Longitudinal Aging Study in India (2020), 8% of Indians are aged 60 years or above today, and this proportion may rise nearly to 19% by the year 2050. Estimates suggest that in India 5.3 million people are currently living with dementia, this figure might be projected to almost triple by the year of 2050.

1.1.1 Importance of Early Detection

Alzheimer's is a degenerative brain disease that is caused by complex brain changes following cell damage. It leads to dementia symptoms that become worse over time and there is still no cure for this disease. So Early Detection is the only way that can help Doctors to slow down the disease progression with Alzheimer's disease.

1.1.2 Neuroimaging Technique

Neuroimaging technique plays an important role in the early detection of Alzheimer's disease. It is a non-invasive technique so images like MRI and PET can capture clear and detailed brain information. MRI is widely used to detect hippocampus atrophy, enlarged ventricles and cortical thinning associated with Alzheimer's disease.

1.1.3 Role of Machine Learning and Deep Learning

Machine Learning and Deep Learning techniques have become increasingly popular for the early detection of Alzheimer's disease using MRI data. Traditional ML methods depend on manual feature extraction, the quality of extracted features and require extensive preprocessing. For all these reasons Deep Learning models outperformed compared to

Traditional ML methods. Deep Learning models are specially used to train on image data as it can learn complex patterns like brain atrophy and structural changes without manual intervention. This study focuses on using MRI data and deep learning techniques for the early detection of Alzheimer's disease.

1.2 Problem Statement

Alzheimer's disease directly affects a person's brain cells, memory and thinking ability. The disease is irreversible and there is no complete cure present at all. Early disease detection and accurate diagnosis are crucial to slow the disease progression. Traditional diagnosis techniques are helpful in the diagnosis of Alzheimer's disease, these techniques are based on clinical analysis and cognitive tests that are often very time-consuming and subjective in nature and also require expert knowledge.

Images such as MRI and PET can provide detailed brain information about brain structure and CSF, also the technique is non-invasive. Analyzing these images manually is a very hard and time consuming process that requires expert knowledge.

In recent years machine learning and deep learning methods have gained so much popularity in the field of image analysis in medical science. These techniques can identify patterns in brain scans, helping medical professionals detect early signs of dementia. But their performance is highly dependent on availability of high-quality and large amounts of datasets and they can become computationally complex.

Therefore, there is a need for developing an automated and reliable system that can analyze MRI data and accurately classify individuals as having AD or CN. This project aims to address this challenge by developing a deep learning-based system to classify between Alzheimer's and normal brains from MRI scans using small amounts of data.

1.3 Objective of the Project

The main objective is to develop a Deep Learning CNN based system to classify between AD vs CN from MRI scan.

The following steps are detailed objective of this project:

1. To collect the MRI data and clinical data from ADNI.
2. To match the MRI data with their clinical records.
3. To construct the label (AD/CN) for MRI scans using clinical records.
4. To apply data cleaning and preprocessing.
5. To divide the dataset into training, validation and testing sets later create a structured dataset folder to move the data.
6. To apply data augmentation on AD training set for handling data imbalance.
7. To load the data from the dataset and prepare it into small batches before model training and testing.
8. To build CNN transfer learning model and train the dataset.
9. To test the model on unseen data and evaluate the performance.
10. To apply Grad-CAM for visualizing the model.

CHAPTER 2: LITERATURE REVIEW

In the following literature review, a general overview is provided about recent studies conducted regarding the use of explainable artificial intelligence, deep learning, transfer learning and MRI for Alzheimer's disease. Some recent advancements in terms of automated diagnoses and classification have been discussed in the reviewed papers. For a better understanding of the current scenario about the detection of Alzheimer's disease, a total of fourteen papers have been considered in this literature review.

An extensive evaluation of the use of deep learning models in the diagnosis of Alzheimer's disease was conducted by Gao and Lima [1]. Different deep learning models, imaging types and diagnosis methods used in the current studies were analyzed in the review. The effectiveness of CNN models in extracting the features related to the disease using neuroimaging images was highly discussed in the paper. However, no specific model design to improve interpretability and classification performance was suggested.

The importance of deep learning approaches for diagnosing and classification of Alzheimer's disease was discussed by Al-Shoukry et al. [2]. Various techniques in terms of machine learning and deep learning have been analyzed by the authors based on MRI and PET imagery data. It was demonstrated that deep learning is increasingly becoming essential in the image processing area in medicine. However, it did not touch upon the issue of explainable AI and advanced transfer learning.

Ekuma et al. [3] proposed an explainable deep learning framework for predicting the quality of MRI images used in the identification of Alzheimer's disease. Three models - DenseNet121, DenseNet169 and Inception-ResNet-v2 were utilized during the study and their predictions were explained using SHAP values. The technique helped identify some important brain areas involved in classifying the image. However, instead of using the binary classification approach to differentiate patients diagnosed with Alzheimer's disease from the control group, the study focused on predicting the severity level at different stages.

Tanveer et al. proposed a framework named Deep Transfer Ensemble (DTE) for Alzheimer's disease classification based on MRI data [4]. To improve the accuracy of classification the proposed methodology utilized ensemble learning, random search and transfer learning techniques. The model highlighted the significance of combining different deep neural networks and it demonstrated an effective performance with high classification accuracy. However, the proposed architecture did not provide any justification for its working process.

Through MRI images, Dardouri [5] developed an approach to detect early stages of Alzheimer's disease based on deep convolutional neural networks. In order to resolve the issue of class imbalance in addition to improving the accuracy of the classification process, the study made use of data augmentation and cross-validation techniques. The proposed technique demonstrated the ability of CNNs in diagnosing diseases. But, the aspect of interpretability was not taken into consideration.

Goel et al. [6] applied both 2D and 3D CNNs for the analysis of Alzheimer's disease diagnosis. VGG16, DenseNet121, InceptionV3, CNN-LSTM models and BrainNet structures were some of the architectures tested in the study. According to the results obtained, deep learning algorithms can employ MRI imaging to diagnose Alzheimer's disease. Nevertheless, the study does not explore techniques in explainable AI nor designs for ResNet based on transfer learning.

Khan et al. [7] developed an approach to apply transfer learning to Alzheimer's Disease Multiclass Classification via MRI images. The VGG16 and VGG19 models that apply stepwise layer freezing and transfer learning were applied in the study. The four types - Alzheimer's disease, Early Mild Cognitive Impairment, Late Mild Cognitive Impairment and Normal Control - were categorized based on gray matter regions in MRI images. Instead of binary classification this approach was applied to multiclass classification.

A method to classify Alzheimer's disease using MRI scans based on deep learning techniques has been proposed by Sorour et al al. [8]. It evaluated the impact of data augmentation on the classification results and analyzed various deep learning architectures. The results revealed that Alzheimer's disease can be effectively classified through deep learning. Nonetheless, interpretation methods were not discussed in this paper.

Through MRI slices, Stoleru and Iftene [9] performed a comparison study to examine transfer learning models for Alzheimer's disease classification. Effectiveness of pre-processing was assessed by analyzing the effect of different preprocessing algorithms on model performance. Several pre-trained models were also compared. This paper shows that transfer learning can efficiently use visualizations but focuses on model comparison instead.

Ghaffari et al. [10] have developed an entirely automated approach for the classification of Alzheimer's disease through the use of brain MRIs based on transfer learning. Pre-trained deep neural networks were analyzed in the study that performed well on different datasets. The technique generated reliable classification results while minimizing human feature extraction efforts. However, no feature explainability methods were used in the algorithm to identify the brain regions impacting decision-making.

Francis and Verma [11] proposed a deep learning-based model using ResNet18 along with the techniques of attention mechanism and transfer learning to detect Alzheimer's disease. In order to improve feature detection from MRI scans, the proposed model incorporated an attention mechanism with a pre-trained ResNet18. The research highlighted the efficiency of the transfer learning technique using a small number of medical data sets. However, attention mechanisms added to the complexity of the model.

Van Oostveen & De Lange [12] studied the imaging modalities involved in diagnosis and monitoring of Alzheimer's Disease. The different types of imaging modality used clinically, such as MRI and PET scan were discussed in the review paper. Emphasis was placed on the importance of imaging biomarkers in detecting any disease at its early stage. However, the review mostly focused on the imaging modality itself.

A convolutional neural network-based diagnosis of Alzheimer's disease utilizing Fine-Tuning Neighborhood Component Analysis was introduced by Ozaltin [13]. In order to improve the efficiency of the classification process the researcher used feature selection techniques along with MRI features that can be classified easily. The results indicated the importance of utilizing the deep learning approach alongside feature selection. However, the suggested approach did not provide any visual support for the classification process.

Grad-CAM is a popular form of explainable AI that provides localization maps of important features in images that play a key role in making decisions within deep learning models; the concept was initially introduced by Selvaraju et al. [14]. It has found wide application in medical image processing and made it easier to understand the behavior of convolutional neural networks. This project did not focus on identifying Alzheimer's, but instead focused on the explainability methodology itself.

CHAPTER 3: SOFTWARE AND HARDWARE REQUIREMENTS

3.1 Software Requirements

Operating System: Windows 10

Programming Language: Python 3.x

Development Environment: Google Colab / Jupyter Notebook

Deep Learning Framework: PyTorch

Computer Vision Library: Torchvision

Libraries Used:

- NumPy – Numerical operations
- Pandas – Data handling
- os – File and directory handling
- shutil – File copying and directory management
- subprocess – Executing external commands from Python
- Matplotlib – Visualization
- OpenCV – Image processing
- Pillow – Image loading
- scikit-learn – Dataset splitting and evaluation metrics
- NiBabel – Handling NIfTI MRI files

Dataset Source: Alzheimer's Disease Neuroimaging Initiative (ADNI)

File Format Conversion Tool: dcm2niix

3.2 Hardware Requirements

- **Processor:** Intel Core i3 (10th generation)
- **RAM:** 16 GB
- **Storage:** 2 TB
- **Graphics Processing Unit (GPU):** Integrated Intel UHD Graphics
- **System Architecture:** 64-bit

CHAPTER 4: METHODOLOGY

4.1 Overall Workflow

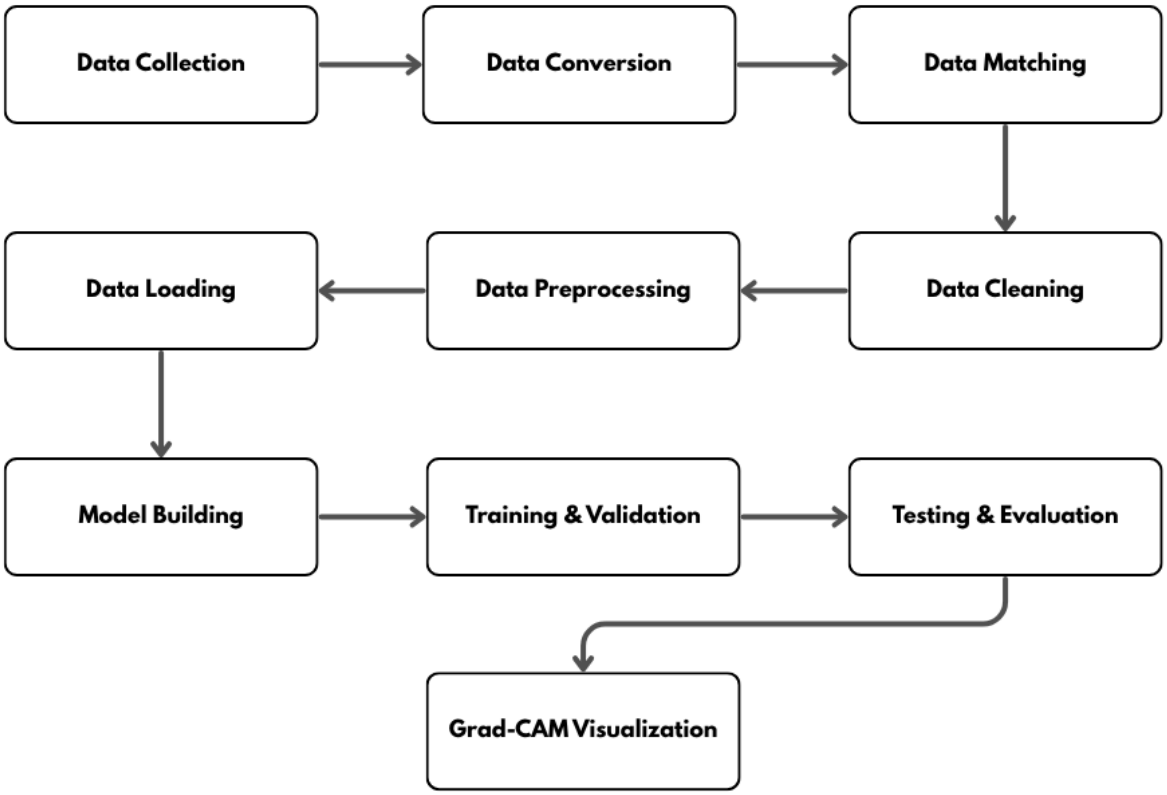


Figure 4-1 Workflow Diagram

4.2 Data Collection

The following two types of data collected for this study:

1. T1 weighted baseline MRI Data of Alzheimer's and Cognitive Normal patients.
2. ADNIMERGE.csv

4.2.1 Data Source

This study required data obtained from Alzheimer's Disease Neuroimaging Initiative (ADNI). The ADNI is a publicly available research database platform that is designed to

develop clinical, imaging, genetic and biochemical biomarkers to study and detect Alzheimer's disease (AD) and its progression.

4.2.2 Dataset Description

- **MRI:** Magnetic resonance imaging (MRI) is a non-invasive medical imaging technique that can provide detailed structural information of the human body including the organs, bones, muscles and blood vessels.
- **T1-Weight:** There are two types of MRI images T1-weighted and T2 weighted. T1 weighted MRI provides a detailed image of brain structure. Where fat appears as bright and fluids like cerebrospinal fluid (CSF) appears as dark. This can become very helpful in study about brain anatomy and spotting any structural changes.
- **Baseline:** Baseline refers to the initial study of a patient. It was later used to compare with future scans of a patient to observe brain changes over time.
- **ADNIMERGE.csv:** ADNIMERGE.csv is a clinical file that contains patient detailed information such as ID, age, gender, diagnosis, visit details and clinical scores.
- **Cognitive Normal (CN):** Cognitive Normal refers to the persons with no memory decline. Whose brain functions normally.
- **Alzheimer's Disease (AD):** Alzheimer's Disease is a brain disorder where a person's brain functions stops working normally and it becomes worse over time.

4.3 Data Conversion

The downloaded MRI data was available in DICOM format which was converted into NifTi format so that data cleaning and preprocessing can be performed easily.

- **DICOM (.dcm):** The DCM file extension is used for Digital Imaging and Communication in Medicine (DICOM). A standard file format that is mostly used by hospitals and MRI/CT scanners to store medical images. A complete MRI scan contains multiple .dcm files. Inside a dcm file it stores the MRI image, patient information and scan details.
- **NifTi (.nii):** Neuroimaging Informatics Technology Initiative (NifTi) stores all the scans combined together in one single file. A single .nii file has all combined image data and metadata. Makes preprocessing easier.

4.4 Data Matching and Class Verification

A total 184 subjects were identified from the obtained MRI dataset. Later matched the MRI subjects with ADNIMERGE clinical file to ensure data consistency and correctness. During the data matching process it was found that in the clinical file there were no records about 11 MRI subjects out of the collected 184 MRI subjects. As for the further study, a total of 173 subjects were considered and remaining 11 subjects were removed from the MRI dataset.

In the next process, verify the total number of subjects that belong to one of the Alzheimer's disease (AD) or Cognitively Normal (CN) categories. To achieve this the classification labels were obtained by matching MRI subject IDs with ADNIMERGE clinical CSV files PTID's. This ensured accurate labeling of each subject. The class distribution of the dataset is as follows:

- Alzheimer's Disease (AD): 36 subjects
- Cognitive Normal (CN): 137 subjects

This distribution indicates the number of subjects available in each category, which is important for understanding about dataset balance and its impact on model performance.

4.5 Data Cleaning

Data cleaning is an essential step to ensure the quality and reliability of the dataset before applying preprocessing and model training. The raw MRI obtained from the ADNI database contained several issues, especially multiple scans for a single subject and 'repeat' scans. Therefore, appropriate cleaning techniques were applied to prepare a consistent dataset.

4.5.1 Removal of Corrupted Files

During the data cleaning process, all MRI files were checked to identify any existence of corrupted and unreadable files. As a result it found that there were no corrupted files present in the MRI dataset. So, no files were needed to remove at this stage. This indicates that the collected dataset was consistent and good quality.

4.5.2 Duplicate Scan Handling

The dataset contained multiple MRI scans for the same subject. These duplicate scans may occur due to the repeated imaging sessions or multiple acquisitions. In the MRI dataset from 173 usable subjects 217 MRI scans were present and there were 40 subjects lists which were found to have multiple MRI scans. To maintain consistency of the dataset, only one MRI scan based on the priority of higher quality was selected for further analysis and remaining scans

were moved from the main dataset to another location. This step helps to avoid bias and redundancy in the dataset.

4.5.3 Handling of REPEAT Scans

The ADNI dataset includes scans labeled as REPEAT which represent repeated acquisitions of the same subject. These scans are not required for this study and may introduce duplication. A total of 57 subjects were identified to have REPEAT scans in the dataset. These scans were selected from the main MRI dataset and shifted to a separate folder. This ensures that the main MRI dataset contains only unique scans for each subject.

After applying the above cleaning steps, each subject of the dataset is represented by a valid single high-quality MRI scan. That ensures the consistency across the dataset. This cleaned dataset is the foundation for further preprocessing and model development steps.

4.6 Data Preprocessing

Data preprocessing techniques are applied on MRI data before feeding the data into the deep learning model. It is a crucial step that ensures data consistency, no data leakage, removes noise from the data and converts high-dimensional data to 2D simple data to remove model complexity. Therefore, several preprocessing techniques were applied to improve data quality and make it suitable for analysis.

4.6.1 Intensity Normalization

MRI scans may have different intensity values due to its various types of equipment and imaging conditions. As a result this makes the model harder to learn consistent patterns from image data. To address this issue, intensity normalization was applied to standardize pixel values across all MRI scans. This ensures that all images have a similar intensity distribution, which helps improve model convergence and performance.

4.6.2 Slice Extraction

In MRI dataset NifTi files contain 3D brain volume data but for models like CNN works better with 2D image data. So, slice extraction was performed to convert 3D MRI voxel data into 2D slices. For each subject 30 middle brain slices were selected because these slices

contain the most relevant anatomical information. These extracted brain slices were resized into 224*224 pixels and saved in PNG image format to later be used as input for the model. This approach reduces computational complexity and increases the amount of training data.

After slice extraction , total image for both CN and AD classes are distributed as:

Total MRI slices	5190
Cognitive Normal (CN)	4110
Alzheimer's Disease (AD)	1080

Table 4-1: Summary of MRI Slice Distribution After Slice Extraction

The result indicates class imbalance in the dataset, where the number of CN samples is significantly higher than AD samples.

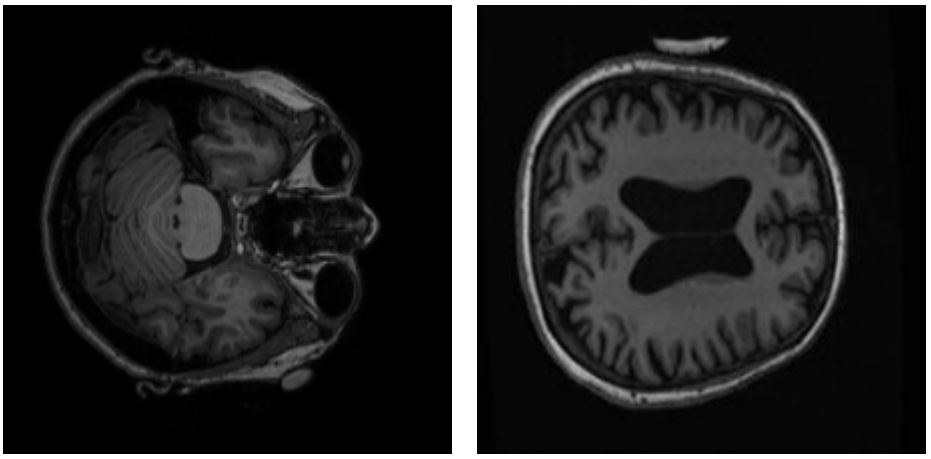


Figure 4-2: MRI Slice Image

4.6.3 DATASET SPLITTING

After performing slice extraction, the MRI dataset was divided into 70% training, 15% validation and 15% testing sets by using a splitting approach for model development and evaluation. This distribution of data is as follows:

Dataset Structure	CN (0)	AD (1)
Training	2880	750
Validation	600	180
Testing	630	150

Table 4-2: Dataset Split Distribution for Training, Validation, and Testing

This distribution ensures that the dataset is properly structured for training and evaluating the deep learning model. It also maintains representation of both classes across all subsets, although a class imbalance is observed with a higher number of CN samples compared to AD.

4.6.4 Data Augmentation

To handle data imbalance, subject-wise data augmentation technique was applied on the AD folder training set. Various techniques like Gaussian noise, random rotations, slight zooming, brightness and contrast adjustments were performed on MRI slices during augmentation. This helps to prevent the model becoming biassed towards CN data. After data augmentation the training dataset of AD increases 750 to 2400.

4.7 Data Loading

Data Loading is the process of loading the data into small batches before feeding them into the model as input. MRI slice images are loaded from the train, validation and test directories using ImageFolder. Before feeding images into the model, the following transformations were applied:

1. Conversion of MRI slice from single from single-channel (grayscale) to 3-channel RGB.
2. Tensor Conversion.
3. Normalize pixel values.

Later load the train, validation and test data separately. Create a dataloader of 16 batches and shuffles training data. Before model training verified the following process:

1. Verify class mapping, batch generation and image dimensions.
2. GPU/CPU setup.

3. Validate the quality of data check error.

4.8 Model Training

4.8.1 CNN

CNN stands for Convolutional Neural Network. It is a deep learning model that was made for performing image processing tasks. It learns important features from images through convolutional layers, pooling layers and activation functions.

- **Convolutional Layer:** In convolution layer a small filter (kernels) has been applied to an input image to detect features such as edges, textures and complex patterns.
- **Pooling Layer:** The pooling layer helps in reducing the dimensions of feature maps produced by the convolutional layer, but at the same time, it retains all the most essential information. It decreases the computational complexity, reduces the spatial dimensions of the data.
- **Activation Function:** Activation function is a mathematical function which determines whether a neuron should be activated or not in a neural network. It brings non-linearity into the model and makes it capable of learning some more complex features from the data.
- **Fully Connected Layer:** The fully connected (FC) layer connects all neurons from the previous layer and takes the responsibility of making the final classification or prediction.

4.8.2 Pretrained ResNet18

Resnet18 has been adopted as pre-trained architecture for this study. The deep convolutional neural network ResNet18 consists of 18 layers and uses residual connections in order to make the training easier and overcome the vanishing gradient issue. Pre-trained weights were loaded into the model based on ImageNet, a dataset with millions of annotated images.

Pre-training allows for using transfer learning, a technique that uses visual features obtained from previously seen images to improve performance in new tasks. The method accelerates the training process, improves its efficiency and yields better results due to a limited number of images. Therefore, the idea was implemented by using a pre-trained ResNet18 network as a feature extractor and classifier of Alzheimer's disease in brain MRI slices.

4.8.3 Model Architecture

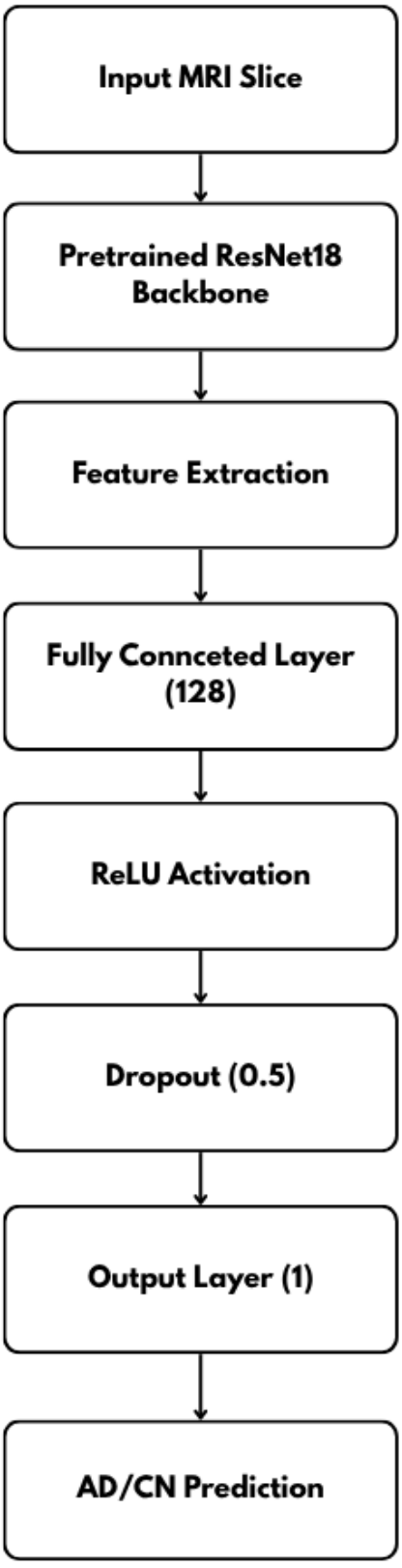


Figure 4-3: ResNet18 Architecture

4.9 Training Strategy

4.9. 1 Transfer Learning Setup

Due to the limitations of MRI data, a pretrained CNN model ResNet18 that was already trained on ImageNet data was built to work in this study. The model is already known to find edges and patterns from image data. The following techniques has been used as model setup and training:

- **Layer Freezing and Fine Tuning:** Layer Freezing means to prevent the model weights from being updated during the training phase. It reduces training time, saves compute resources and prevents overfitting specially for small datasets.
  Fine Tuning is the process where some of the pretrained layers allow himself to update their weights during training. This can help to improve model accuracy.
  In this study both layer freezing and fine tuning techniques have been used in model experiments.

- **Activation Function (ReLU):** An activation function is used in a neural network to decide whether to activate a neuron or not. The use of this function adds non-linearity into the model to make it possible for the learning of complicated patterns.
  The Rectified Linear Unit (ReLU) acts as the activation function; if the input is positive then the output value is the same as input otherwise it produces zero if it is negative.
  Formula: $f(x) = max(0, x)$

- **Dropout Regularization:** To solve the overfitting problem, dropout was set at 0.5. This means that 50 percent of the neurons will randomly deactivate during training.

- **BCEWithLogitsLoss:** BCEWithLogitsLoss is a loss function which can be applied in cases when there is a need to perform binary classification. Thus only two classes are predicted as the output value. This loss combines the sigmoid layer with binary cross entropy loss function.

- **Adam Optimization Algorithm:** The Adam optimization algorithm was used in the tuning of the network weights. Adam uses the concept of adaptive learning rates with momentum optimization for increasing the efficiency of the training process. The use

of a low learning rate may result in slow training, but this increases accuracy and makes training more stable.

- **Learning Rate:** A learning rate refers to a hyperparameter responsible for weight updates in the training process. It represents how far the optimizer moves towards achieving convergence when minimizing the loss function. A high learning rate ensures quick training; however, it can cause the optimizer to miss the global minima.

- **Weight Decay:** Weight decay refers to the application of a penalty in order to prevent overfitting through regularizing large weight values.

- **Epochs and Early Stopping:** One epoch represents passing through the whole training data set once through the artificial neural network. The algorithm applies weighted updates from all training data in each epoch to improve its efficiency. Early stopping defined with the help of the patience parameter was used to avoid unnecessary training and stop the training process at a point when there was no improvement.

- **Model Saving Strategy:** The model was saved each time a better model was obtained based on a lower validation loss.

4.10 Validation Procedure

Validation procedure was performed at the end of every training epoch, in order to evaluate the performance of the model during training. Validation loss was used as the key metric for the evaluation of the generalization of the model. In order to avoid overfitting and excessive training time, an early stopping approach was implemented. The training would be stopped automatically, once no improvement was observed in the validation loss for a certain amount of epochs (patience).

CHAPTER 5: EXPERIMENTAL SETUP

5.1 Experimental Design

In this study multiple experiments were conducted using different datasets and training settings to analyze model performance. Three models were customized by conducting multiple experiments to analyze the impact of data augmentation, transfer learning strategies and hyperparameter configurations on Alzheimer's Disease classification performance.

5.1.1 Model 1

For the model 1 real MRI dataset with no augmentation was used to analyze the model performance. Following hyperparameters configurations was applied during training experiments:

- Freeze Backbone layer or Unfreeze 4th layer.
- Learning Rate 1e-4 or 1e-5.
- Number of epochs 25 and patience 5.
- Save the best model during the training phase.

5.1.2 Model 2

For the model 2 augmented MRI dataset was used to analyze the model performance. Following hyperparameters configurations was applied during training experiments:

- Freeze Backbone layer or Unfreeze 4th layer.
- Learning Rate 1e-4 or 1e-5.
- Number of epochs 25 and patience 5.
- Save the best model during the training phase.

5.1.3 Model 3

Model 3 is the enhanced experiment of model 2 where the same augmented dataset was used but as the expectations of a better improved model performance new configuration and

threshold tuning were added during the experiment. Following hyperparameters configurations was applied during experiments:

- Freeze Backbone layer or Unfreeze 4th layer.
- Learning Rate 1e-4/1e-5 and Weight Decay 1e-5.
- Number of epochs 25 and patience 5.
- Save the best model during the training phase.
- Apply threshold tuning.

CHAPTER 6: RESULTS AND EVALUATION

6.1 Testing Procedure

Subject-level evaluation was performed by grouping MRI slices belonging to the same subject, generating prediction probabilities for each slice using the trained pretrained ResNet18 model, averaging these probabilities across all slices of a subject and assigning a final class label based on the mean probability. Performance was then assessed using Accuracy, Recall, Precision, F1-Score, AUC and Confusion Matrix metrics.

6.2 Performance Metrics

To evaluate the classification performance of the proposed model, several standard metrics were used, including Accuracy, Precision, Recall, F1-Score, AUC and Confusion Matrix.

- **Accuracy:** Accuracy refers to how much the model makes correct predictions out of total predictions.

$$Accuracy = (TP + TN)/(TP + TN + FP + FN)$$

- **Precision:** Precision measures how many predicted positive cases the model makes are actually positive.

$$Precision = TP/(TP + FP)$$

- **Recall:** Recall measures how much percentage of all positive cases were correctly predicted as positive.

$$Recall = TP/(TP + FN)$$

- **F1-Score:** The F1 score balances between precision and recall using their harmonic mean.

$$F1 - Score = (2 * Precision + Recall)/(Precision + Recall)$$

- **AUC:** The Area under the curve (AUC) is used to determine the model's performance to differentiate between the classes.

$$\text{AUC}=\int_0^1 TPR(FPR)d(FPR)$$

Where,

$$\text{TPR} = \frac{TP}{TP+FN}$$

$$\text{FPR} = \frac{FP}{FP+TN}$$

- **Confusion matrix:** A confusion matrix is a measurement tool used to test how well a classification model works by measuring the number of true positives, true negatives, false positives, and false negatives.

	Predicted Positive	Predicted Negative
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

Table 6-1: Binary Classification Confusion Matrix

$Confusion\ Matrix$: $\begin{bmatrix} TP & FN \\ FP & TN \end{bmatrix}$

6.3 Experiment Results

Model No	Experiment No	Techniques Used	Accuracy	Recall	Precision	F1-Score	AUC	C.M	
Model 1	Experiment 1	Freeze Backbone Layers	84.62	40	66.67	50	91.43	20	1
		lr = 1e-4						2	3
	Experiment 2	Freeze Backbone Layers	80.77	0	0	0	89.52	21	0
		lr = 1e-4						5	0
		Load the Best Model							
	Experiment 3	Unfreeze Layer 4	76.77	40	40	40	84.76	18	3
		lr = 1e-4						3	2
		Load the Best Model							

	Experiment 4	Unfreeze Layer 4	80.77	0	0	0	86.67	21	0
		lr = 1e-5						5	0
		Load the Best Model							
	Experiment 5	lr = 1e-5	80.77	0	0	0	77.14	21	0
		Load the Best Model						5	0
	Experiment 6	Unfreeze Layer 4	76.92	20	33.33	25	85.71	19	2
		lr = 1e-4						4	1

Table 6-2: Model 1 All Experiment Results

Model No	**Experiment No**	**Techniques Used**	**Accuracy**	**Recall**	**Precision**	**F1-Score**	**AUC**	**C.M**	
Model 2	Experiment 1	Freeze Backbone Layer	88.46	80	66.67	72.73	92.38	19	2
		lr = 1e-4						1	4
	Experiment 2	Freeze Backbone Layer	84.62	60	60	60	82.86	19	2
		lr = 1e-4						2	3
		Load the Best Model							
	Experiment 3	Freeze Backbone Layer	84.62	60	60	60	79.05	19	2
		lr = 1e-5						2	3
	Experiment 4	Unfreeze Layer 4	80.77	40	50	44.44	85.71	19	2
		lr = 1e-4						3	2
	Experiment 5	Unfreeze Layer 4	84.62	60	60	60	86.67	19	2
		lr = 1e-5						2	3

Table 6-3: Model 2 All Experiment Results

Model No	Experiment No	Techniques Used	Accuracy	Recall	Precision	F1-Score	AUC	C.M	
Model 3	Experiment 1	Unfreeze Layer 4	69.23	60	33.33	42.86	75.24	15	6
		lr = 1e-4						2	3
		weight decay = 1e-5							
		Threshold Tunning							
	Experiment 2	Freeze Backbone Layers	50	80	25	38.1	79.05	9	12
		lr = 1e-4						1	4
		weight decay = 1e-5							
		Threshold Tunning							
	Experiment 3	Freeze Backbone Layers	50	80	25	38.1	80.95	9	12
		lr = 1e-4						1	4
		Threshold Tunning							
	Experiment 4	Freeze Backbone Layers	88.46	80	66.67	72.73	87.62	19	2
		lr = 1e-4						1	4
		weight decay = 1e-5							
	Experiment 5	Unfreeze Layer 4	76.92	40	40	40	80	18	3

		lr = 1e-4						3	2
		weight decay = 1e-5							

Table 6-3: Model 3 All Experiment Result

6.4 Comparative Analysis of All Experiments

6.4.1 Dataset Comparison (Real vs Augmented)

Dataset	Best Experiment	Accuracy	Recall	Precision	F1	AUC
Real Dataset (Model 1)	Exp 1	84.62%	40.00%	66.67%	50.00%	91.43%
Augmented Dataset (Model 2)	Exp 1	88.46%	80.00%	66.67%	72.73%	92.38%

Table 6-4: Comparison Results of Real and Augmented Datasets

6.4.2 Freezing vs Fine-Tuning Comparison

Strategy	Experiment	Accuracy	Recall	F1	AUC
Freeze Backbone	Model 2 Exp 1	88.46%	80.00 %	72.73%	92.38 %
Unfreeze Layer 4	Model 2 Exp 5	84.62%	60.00 %	60.00%	86.67 %

Table 6-5: Performance Comparison of Freeze Backbone and Unfreeze Layer 4 Strategies

6.7.3 Learning Rate Comparison

Learning Rate	Best Accuracy	Best Recall	Best F1
1e-4	88.46%	80.00%	72.73%
1e-5	84.62%	60.00%	60.00%

Table 6-6: Performance Comparison of lr 1e-4 and 1e-5

6.7.4 Threshold Comparison

Experiment No	Threshold value	Accuracy	Recall	Precision	F1-Score	AUC	C.M	
Experiment 1	0.3	73.08	40	33.33	36.36	82.86	17	4
							3	2
Experiment 2	0.35	76.92	40	40	40	82.86	18	3
							3	2

Experiment 3	0.4	76.92	40	40	40	82.86	18	3
							3	2
Experiment 4	0.45	73.08	20	25	22.22	82.86	18	3
							4	1

Table 6-7: Performance Comparison of Threshold Tuning

6.4.5 Overall Best Model Selection

Model	Experiment No	Techniques Used	Accuracy	Recall	Precision	F1-Score	AUC	C.M	
Model 2	Experiment 1	Freeze Backbone Layer	88.46	80	66.67	72.73	89.52	19	2
		lr = 1e-4						1	4

Table 6-8: Overall Best Model Performance Result

CHAPTER 7: MODEL EXPLAINABILITY

7.1 Introduction to Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is a technique used for visualizing the output generated by a CNN and interpreting the basis of its decisions. It identifies the parts of the image whose activation contributed significantly to the decision made. Grad-CAM computes the heat map of the activated image by taking the gradient of the target class through the final convolutional layer. In this work, Grad-CAM will be used to visualize the brain regions that impacted the decision making for AD and CN patients.

7.2 Grad-CAM Implementation

After model training, Grad-CAM was applied to visualize the areas of MRI slices that influenced the model's predictions. The model was first switched to evaluation mode to ensure consistent inference. Target layer for activation map generation was Layer4[-1].conv2 of the ResNet18 network structure. In order to extract feature maps from the target layer, a forward pass was done through the network, followed by obtaining gradient values related to the prediction of the particular class using a backward pass. Channel-wise importance scores were calculated by applying global averages and heatmaps were created. The heatmap

Obtained was normalized and plotted over the original MRI image. High activations indicate areas with significant impact on the classification process.

7.3 Visualization Results

Subject-wise Grad-CAM visualizations were generated for correctly classified subjects from both CN and AD categories. Images belonging to the same subject were grouped together using the subject identifier extracted from the image filename, For each subject, five representative MRI slices were selected from different positions across the brain volume. The original MRI slices and their corresponding Grad-CAM visualizations were displayed for qualitative analysis.

7.3.1 Alzheimer's Disease Subjects

In the subjects with Alzheimer's disease, the Grad-CAM technique brought into
focus on certain anatomical structures of the brain that greatly helped make the classification decision by the model. From the findings made from activation maps, it is clear that there was attention to specific anatomical structures in the brain; hence, there was attention to structure during diagnosis of Alzheimer's disease.

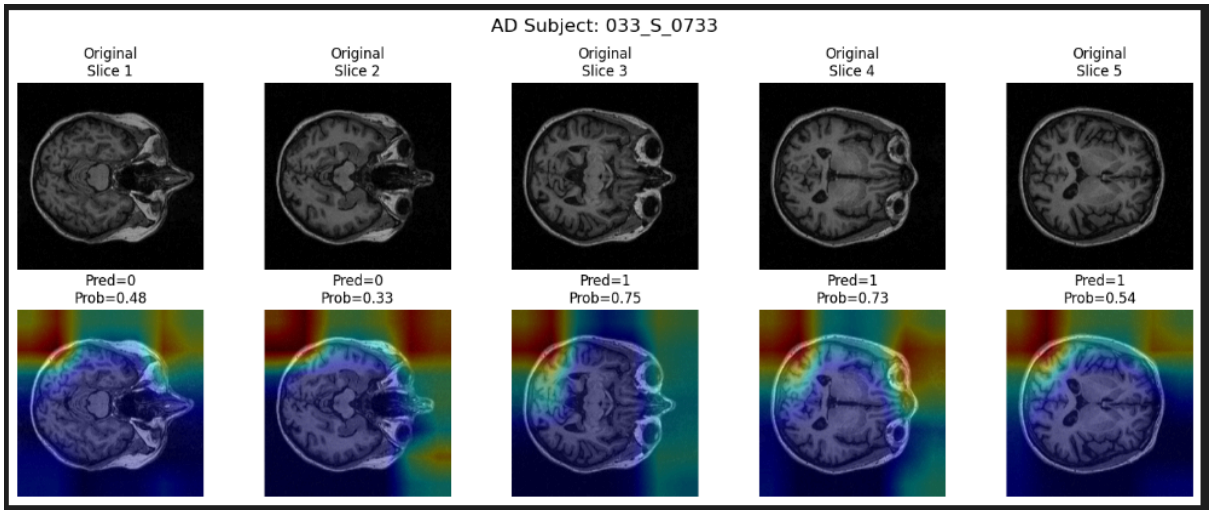


Figure 7-1: Grad-CAM Visualization for AD subject

7.3.2 Cognitive Normal Subjects

For the Cognitive Normal subjects, Grad-CAM generated activation maps that differed from those observed in Alzheimer's Disease subjects. The areas of interest were those which exhibited anatomical properties of the brain. In other words, these were healthy parts since

there was no presence of any diseases or defects. Thus, the model accurately distinguished them based on the brain's areas which were free from detection.

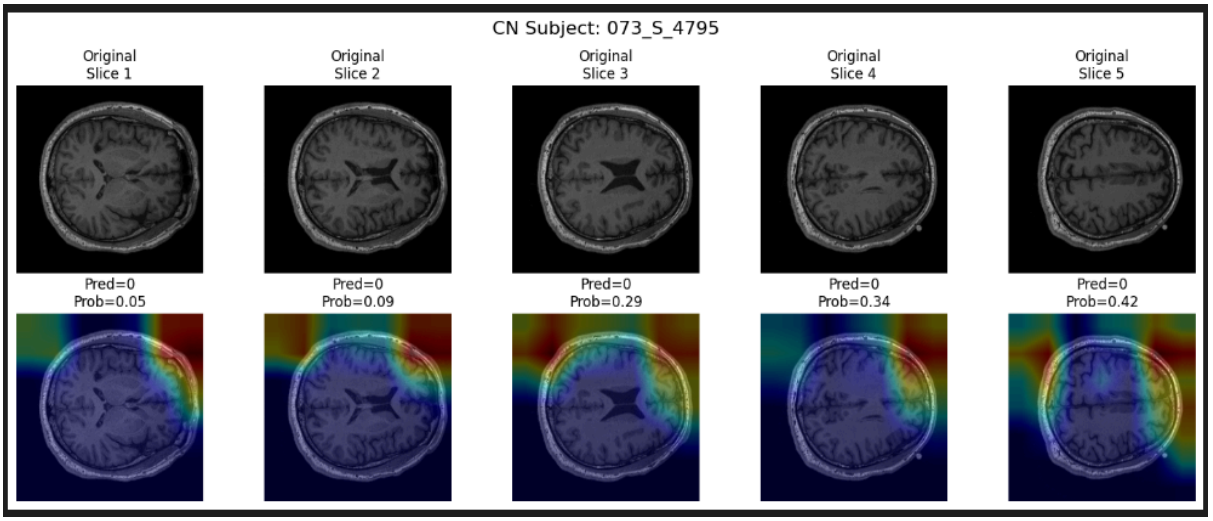


Figure 7-1: Grad-CAM Visualization for CN subject

7.4 Interpretation of Attention Maps

The Grad-CAM visualizations make it possible to understand how the decision process takes place. The regions shown in brighter colors such as red or yellow have higher significance in terms of classification. Cooler colors show low relevance. Therefore, the attention maps show that the model used significant brain regions instead of arbitrary areas of the images.

CHAPTER 8: DISCUSSION

8.1 Key Findings

The performance of a transfer learning-based ResNet18 model, the classification of AD and CN subjects based on MRI data was evaluated in this study. The results demonstrate that experiments on augmented MRI dataset can significantly improve the model performance compared to experiments on real MRI dataset.

The comparison of layer freezing and fine-tunning strategies showed that frozen backbone layer configuration helps to achieve better results over fine tuning of the fourth residual block. Validation phase the loss for every epoch becomes higher when fine-tune strategy applied on the model compared with layer freezing strategy. Moreover, a learning rate value of 1e-4 was found to be superior to 1e-5, thus achieving a higher recall and F1-score.

Furthermore, weight decay and threshold tuning were used to improve the ability of models to generalize during classification tasks. The best model from all the experiments was frozen ResNet18 backbone with the learning rate of 1e-4 trained on augmented datasets giving an accuracy of 88.46% and f1-score 72.73%.

The results indicate that transfer learning with data augmentation is a suitable method for MRI-based classification of Alzheimer's Disease when the dataset is limited and imbalanced.

8.2 Strengths of the proposed model approach

The proposed approach has several strengths for Alzheimer's Disease classification using MRI data. First the use of transfer learning with ResNet18 was a very efficient way of performing feature extraction, which reduced the model computational cost and saved time during training. Moreover, such an approach made it possible to study and work with small datasets.

Second data augmentation techniques increase the MRI slices samples on training AD set to maintain balance between CN and AD category and early stopping helped prevent the model becoming overfit by stop training when the validation loss no longer improved.

Another strength is the use of subject-level evaluation approach, it gives more accurate assessment compared to evaluating every MRI slice individually. Moreover, using the Grad-CAM visualization approach helped in the interpretation of the results by identifying the brain regions that aided in the classification process.

In summary, the introduced method is straightforward, computationally inexpensive, and can attain high classification accuracy using restricted MRI scans.

8.3 Limitations

Despite achieving promising results, there are several limitations that need to be addressed. First, the size of the dataset was relatively very small. A larger dataset can enhance the robustness and generalization ability of the model.

Second, In the dataset number of MRI slices belonging to AD class was very small compared to CN class because of the very few AD patients this leads to data imbalance. Even though data augmentation was applied on AD training set to increase the MRI samples as a solution, it does not fully capture the diverse features in MRI data.

Another limitation is that 2D brain MRI slices were extracted from 3D brain scans. However this approach reduces the computational complexity but on the other hand there might be the possibility of losing some important spatial information.

In addition, the model was built only to perform binary classification between AD and CN subjects. The other phases like Mild Cognitive Impairment (MCI) were ignored. Additionally, only the ResNet18 model was tested and no comparison with other deep learning models was made.

CHAPTER 9: CONCLUSION

9.1 Conclusion

This study refers to a deep learning based approach to detect Alzheimer's Disease from classifying between Cognitive Normal (CN) and Alzheimer's Disease (AD) using MRI data. Required MRI scans and clinical information for this study was collected from ADNI. Before training and evaluation the dataset has passed through multiple stages like data matching, conversion, cleaning and preprocessing technique. A pretrained ResNet18 model has been selected for classification.

Several tests have been conducted to test the effect of the following on the accuracy of the models: data augmentation, layer freezing, fine-tuning, learning rates, weight decay and thresholds. The results showed that data augmentation is what had the largest effect and that a frozen backbone combined with a learning rate of 1e-4 worked best. Subject-wise analysis

and Grad-CAM visualization confirmed that the suggested solution was robust and interpretable.

Overall the study proved that using transfer learning strategy with data augmentation is an effective way to classify Alzheimer's Disease, even working with limited and imbalanced datasets.

9.2 Future Enhancement

There exist many areas that could be improved in the feature for better enhancement in the performance and applications of the suggested system.

- To improve model performance and generalization use a large MRI dataset.
- Improve model from binary classification to multi classification task by MCI stage.
- Use M06, M12 and M24 MRI scans to study disease progression over time.
- Use combined MRI and PET scan data as they give more detailed information related to brain structures and functions.
- Test the dataset on other deep learning models such as ImageNet, DenseNet and EfficientNet etc.
- Work with 3D data to study more about Alzheimer's Disease.
- Utilize the created model to deploy it as a web or a desktop app to facilitate MRI predictions.

The above suggestions can help in building an improved system that can be used for the diagnosis of Alzheimer's Disease.

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